

Influence of Demographic and Clinical Factors on Ovarian Cancer

Survival in the United States

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Abstract

Ovarian cancer survival varies widely among women and is influenced by both disease severity and patient characteristics. Prior population-based studies demonstrate that older age, advanced stage at diagnosis, and limited access to specialized care are linked to higher risk of mortality among ovarian cancer patients (Urban et al., 2016). This study analyzes the impact of demographic and clinical factors, including age at diagnosis, race/ethnicity, grade, and stage on ovarian cancer survival in the United States using Kaplan-Meier survival analyses. The most influential predictor of survival was stage at diagnosis as it had substantially poorer outcomes seen in patients with distant-stage disease. Across racial groups survival also declined as the patients increased in age. Findings from research studies of rare, aggressive ovarian cancer subtypes such as ovarian carcinosarcoma spotlights the role of tumor stage and coexisting conditions, with advanced stage disease linked to a higher risk of mortality (Chang et al., 2025). These findings emphasize the significance of early detection, disease severity, and access to effective treatment in improving ovarian cancer survival.

Introduction

Ovarian cancer remains one of the leading causes of cancer related deaths among women and it is a major public health concern in the United States (Siegel et al., 2024; Torre et al., 2018). As treatment advances, survival outcomes are limited, particularly for patients diagnosed at advanced stages (Lheureux et al., 2019). Through population based research, studies have shown that mortality is especially high directly after diagnosis, approximately 25% of patients with advanced ovarian cancer die within 90 days of diagnosis, and over 40% die within the first year (Urban et al., 2016). Demographic and clinical factors influence survival rates including but not limited to

age at diagnosis, race/ethnicity, and stage at diagnosis. Older age, advanced disease, and limited access to specialized care have been associated with worse outcomes (Urban et al., 2016; Hildebrand et al., 2019), while treatment such as chemotherapy has been shown to improve survival (Lheureux et al., 2019). Having knowledge or a sense of understanding of how these factors impact survival is essential for identifying disparities and improving early detection, treatment, and resource allocation. This study examines how demographic and clinical characteristics influence ovarian cancer survival in the United States using population based data. The study hypothesized that advanced stage at diagnosis, older age, higher tumor grade, and Black race would be associated with poorer survival outcomes, while earlier-stage disease and Asian or Pacific Islander race would be associated with improved survival.

Methods

This was a population-based survival analysis using data obtained from the Surveillance, Epidemiology, and End Results (SEER) database through SEER*Stat software. The analysis included female patients diagnosed with malignant ovarian cancer between 2000 and 2022. Demographic variables included age at diagnosis and race/ethnicity (White, Black, American Indian/Alaska Native, Asian or Pacific Islander), while clinical variables included stage at diagnosis (localized, regional, and distant) and tumor grade. The outcome was overall survival, measured from diagnosis until last follow up or passing.

Kaplan-Meier survival analyses were performed to estimate survival probabilities at 1, 3, and 5 years. Survival tables were produced by age group (60+), race/ethnicity, and stage at diagnosis. In addition, frequency statistics were used to produce descriptive statistics and grouped bar charts

showing the distribution of cases by age group and race within each stage category. Survival estimates were calculated with 95% confidence intervals. The analyses were used to analyze differences in survival outcomes across demographic and clinical subgroups.

A multivariable Cox proportional hazards regression analysis was used to identify independent predictors of ovarian cancer survival using SEER*Stat survival data and analyzed in RStudio. The model determined the association between demographic and clinical variables and risk of mortality among ovarian cancer patients. The variables that were included in the analysis are age at diagnosis, race/ethnicity, stage at diagnosis, and tumor grade. Hazard ratios (HRs) with 95% confidence intervals (CIs) were calculated to estimate the relative risk of death associated with each variable. The comparison groups that were used are localized stage disease, White race, and the reference age and grade categories. Wald chi-square test was used to determine statistical significance, with a significance level of $p < 0.05$. Model performance was assessed using likelihood ratio. The Cox proportional hazards model was used to determine whether demographic and clinical characteristics independently predicted ovarian cancer survival.

Table 1. Patient Characteristics

Variable	n (%)
Age Group	
0-19 years	1677 (1.3%)
20-39 years	8371 (6.6%)
40-44 years	5910 (4.7%)
45-49 years	9433 (7.5%)
50-54 years	12813 (10.2%)
55-59 years	14577 (11.6%)
60-64 years	15176 (12%)
65-69 years	14321 (11.4%)
70-74 years	13308 (10.6%)
75-79 years	11622 (9.2%)
80-84 years	9500 (7.5%)
85-89 years	6127 (4.9%)
90+ years	3267 (2.6%)
Race/Ethnicity	
White	103228 (82.3%)
Black	10446 (8.3%)
American Indian/Alaska Native	935 (0.7%)
Asian or Pacific Islander	10883 (8.7%)
Stage at Diagnosis	
Localized	14904 (16.3%)
Regional	18894 (20.6%)
Distant	57845 (63.1%)
Tumor Grade	
Grade I	6810 (6.8%)
Grade II	11601 (11.6%)
Grade III	28362 (28.5%)
Grade IV	13434 (13.5%)
Unknown	39472 (39.6%)

Figure 1. Kaplan–Meier Survival Curve by Stage

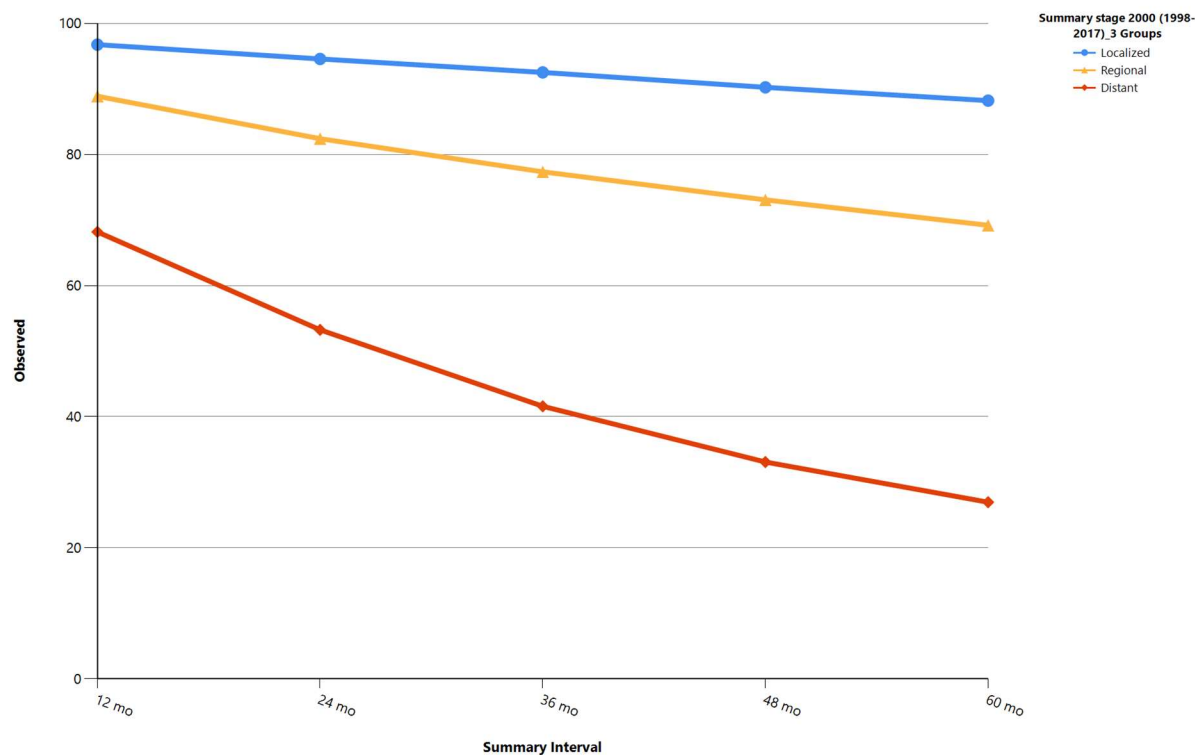


Figure 2. Kaplan–Meier Survival Curve by Age Group

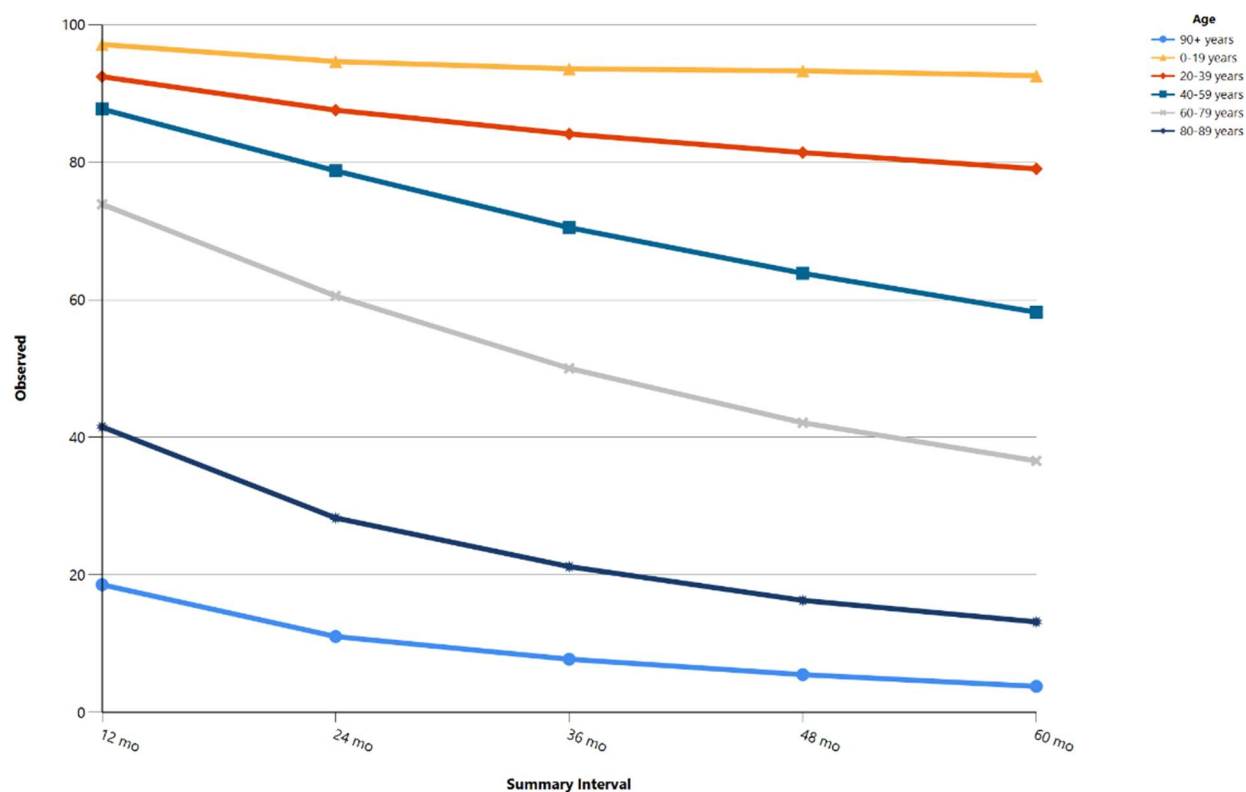
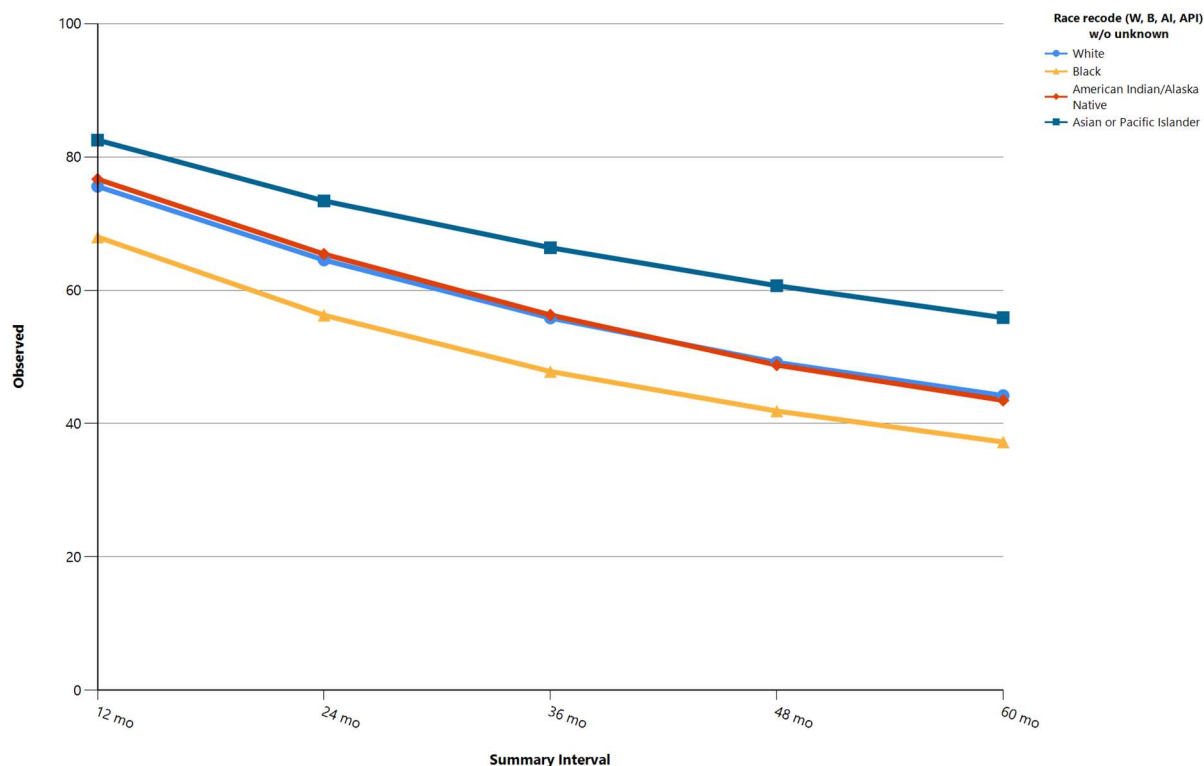


Figure 3. Kaplan–Meier Survival Curve by Race/Ethnicity



Results

Overall survival outcomes differed by stage at diagnosis, grade, age, and race/ethnicity. Patients diagnosed with localized ovarian cancer had the highest survival, with approximately 89% surviving at 5 years compared to 71% for regional disease and only 28% for distant-stage disease. As age increased, survival declined consistently. Patients aged 60–64 had a 5-year survival of approximately 45%. However, those aged 85–89 and 90+ had lower outcomes of about 10% and 4%, respectively. In this study, racial differences in overall survival were also observed. White patients had a higher 5-year observed survival of approximately 45%, while Black patients had around 39% survival rates. However, higher results were demonstrated in the Asian and Pacific

Islander patients with approximately 57% surviving at 5 years. Descriptive analysis demonstrated that most cases were diagnosed at a distant stage and were more common in older age groups, particularly among White patients. Overall, these findings suggest that later stage diagnosis, older age, and certain racial groups are associated with lower ovarian cancer survival.

To further assess these differences and identify independent predictors of survival, a multivariable Cox proportional hazards regression model was conducted including age, race/ethnicity, stage at diagnosis, and tumor grade. The analysis included 124,215 patients with 77,935 events and was highly statistically significant overall (Likelihood Ratio test $\chi^2 = 57,653$, Wald test $\chi^2 = 49,416$, Score test $\chi^2 = 61,804$; all $p < 0.001$), with good model discrimination (concordance = 0.746). Stage at diagnosis appeared as the strongest predictor of mortality. Compared to patients with localized disease, patients with regional disease had a higher risk of death (HR = 2.87, 95% CI: 2.39-3.45, $p < 0.001$). Patients with distant-stage disease experienced a higher risk of death compared to those diagnosed at a localized stage (HR = 8.03, 95% CI: 6.69-9.64, $p < 0.001$). As age increased, the risk of mortality also increased, particularly within the elderly age groups, with patients aged 80-84 (HR = 4.55, $p = 0.032$), 85-89 (HR = 6.55, $p < 0.001$), and 90 years or older (HR = 10.17, $p < 0.001$). These results demonstrate the elevated risk of death compared to the reference group. There were significant racial disparities which identified Black patients with a higher risk of death compared to White patients (HR = 1.19, 95% CI: 1.10-1.30, $p < 0.001$). While Asian or Pacific Islander patients had a lower risk of death (HR = 0.78, 95% CI: 0.71-0.85, $p < 0.001$). Tumor grade was also another strong predictor of survival, with poorly differentiated tumors (Grade 3) associated with increased mortality (HR = 4.88, $p < 0.001$) and undifferentiated tumors (Grade 4) showing even higher risk (HR = 7.80, $p < 0.001$). Overall, these findings

suggest that stage at diagnosis is the most influential predictor of ovarian cancer survival, followed by tumor grade and age, with race/ethnicity also contributing to differences in outcomes.

Table 2: Cox Regression Results

Variable	Hazard Ratio (HR)	95% CI	p-value
Age Group			
01–04 years	Reference	-	-
05–09 years	0.14	0.03–0.64	0.012
10–14 years	0.14	0.03–0.73	0.006
15–19 years	0.28	0.07–1.13	0.073
20–24 years	0.46	0.14–1.85	0.272
25–29 years	0.5	0.12–2.02	0.331
30–34 years	0.73	0.21–2.94	0.662
35–39 years	0.85	0.21–3.39	0.813
40–44 years	1.02	0.41–2.57	0.973
45–49 years	1.16	0.29–4.62	0.838
50–54 years	1.35	0.34–5.40	0.672
55–59 years	1.51	0.38–6.02	0.564
60–64 years	1.73	0.43–6.93	0.437
65–69 years	2.05	0.51–8.21	0.309
70–74 years	2.48	0.62–9.93	0.198
75–79 years	3.25	0.81–13.01	0.095
80–84 years	4.55	1.14–18.19	0.032*
85–89 years	6.55	1.64–26.02	0.008**
90+ years	10.17	2.54–40.67	0.001**
Race/Ethnicity			
Asian/Pacific Islander	0.78	0.72–0.85	<0.001***
Black	1.19	1.06–1.34	<0.001***
White	Reference	-	-
Stage at Diagnosis			
Localized	Reference	-	-
Regional	2.87	2.39–3.45	<0.001***
Distant	8.03	6.69–9.64	<0.001***
Tumor Grade			
Grade 1	Reference	-	-
Grade II	2.07	1.66–2.59	<0.001***
Grade III	4.88	4.01–5.93	<0.001***
Grade IV	7.8	6.49–9.38	<0.001***
Note: p < 0.05 (*), p < 0.01 (**), p < 0.001 (***)			

Discussion

This study demonstrates that both clinical and demographic factors significantly contribute to ovarian cancer survival in the United States. Both demographic and clinical factors contributed to ovarian cancer survival outcomes, although clinical characteristics appeared to be the strongest predictors. From the variables that were analyzed, stage at diagnosis was the strongest independent predictor of mortality. Tumor grade was also a major factor of survival, with higher grade tumors associated with increased mortality, reflecting the aggressive nature of high-grade disease (Kurman & Shih, 2016). Patients aged 80 years and older, experienced worse outcomes of survival, which may reflect differences in overall health status, treatment eligibility, or disease burden. Underlying health conditions and concurrent illnesses can also further impact survival outcomes and complicate treatment decisions and results.

Even though racial differences were observed, with Black patients experiencing higher mortality and Asian or Pacific Islander patients showing improved survival compared to White patients. These findings summarize the disparities in access to timely diagnosis and high-quality treatment (Hildebrand et al., 2019; Terplan et al., 2012). Compared to stage and tumor grade, racial differences showed more moderate effects on survival. The research supported the hypothesis, which proposed that advanced stage at diagnosis, older age, higher tumor grade, and Black women would be associated with an increase in mortality. On the other hand, earlier stages of diagnosis and Asian or Pacific Islander race would be associated with increased survival outcomes. Through the Cox proportional hazards analysis, we can confirm that these factors are independent predictors of survival rather than just descriptive statistics. Overall, these results demonstrate that early detection is critical since survival declines with more advanced disease

(Menon et al., 2021). Improving ovarian cancer survival outcomes requires not only earlier diagnosis but also ensuring access to high-quality cancer care and treatment.

This study has several limitations. The SEER database does not include detailed information on chemotherapy regimens, recurrence, socioeconomic factors, or comorbidities, which could influence survival outcomes. Despite this limitation, the population-based sample provides valuable insight into demographic and clinical predictors of ovarian cancer survival in the United States.

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